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Assessment of Post-Financial Crisis European Bank Mergers and Acquisitions on Shareholder Value

Abstract

This paper studies creation of shareholder value for post-financial crisis mergers and acquisitions among European banks. It uses pre-existing frameworks from prior transactions' research to analyse recent deals and produce results with greater relevance to the post-crisis financial industry.

The paper uses an event study approach for analysis. These results are evaluated on a set of hypotheses to conclude whether value was created for shareholders of acquiring banks, and whether particular deal characteristics affect results. It finds that a majority of transactions create value for acquiring shareholders. It also finds that domestic and larger deals create more value, however, fails to establish any significant relationship with returns.

Some findings are in line with the pre-crisis literature with exceptions for other deal characteristics. These findings are judged to be a factor of the changing motivations for deals and how investors evaluate the ability for transactions to create value given new priorities.

Introduction

European banks' commercial environment of operations has been fundamentally altered since the 2008 financial crisis. The average return on capital for Europe's lenders had fallen to 6.7% in 2020, less than half of the 15% in 2007 (Stelmach et al., 2021). The value of the STOXX Euro 600 Banks index that tracks the performance of the continent's largest listed banks had fallen nearly 80% between 2007 and 2020 to values comparable to 1990. This contrasts with the performance of its American counterpart – the Dow Jones US bank index - that had fully recovered lost market capitalisation and set a new high coming into 2021.

The divergence among regions has been explained due to the process of recapitalisation. Homar and Winjbergen (2016) argue that recapitalised banks recover quicker from a financial crisis. US banks received a TARP-style injection of liquidity and subsequently raised equity – recovering quicker from the crisis. European banks didn't engage in similar equity raises following the financial crisis (Benink and Huizinga, 2013) and were hit by the Eurozone debt crisis in 2010-11 that destabilised even relatively healthy banking sectors such as those in Greece (Hau, 2011).

In addition to the post-crisis recovery, a low-growth, low interest rate environment has hampered profitability. Return on equity has been forecasted to remain under 6% through till 2025. This is continuation of a longer-term trend where returns from European banks have fallen below their respective costs of equity since 2008 (Stelmach et al., 2021). The phenomenon has been amplified as common equity tier 1 ratio for EU banks was pushed up to an average of 13% in 2020 (Norrestad, 2021). High capital requirements alongside falling returns have resulted in banks trading below their book values (Grodzicki et al., 2019), and in some cases with capital in excess of their market capitalisations (Latorre, 2021).

The change in the underlying fundamentals of the industry – particularly greater capital requirements, and lower profitability and growth – have created an excess of financial infrastructure causing overcapacity in European banks (Magnus et al., 2017). Since these well capitalised institutions are unlikely to fail, M&A has to be studied as an alternative form of exit from the market. The European banking sector has been under vast reorganisation since the financial crisis. In the immediate aftermath of the crisis, mergers and acquisitions activity was driven by the need to shore up liquidity and prevent banks from collapsing. This activity continued going into the Eurozone debt crisis.

As market valuations dropped significantly and asset quality deteriorated, M&A activity has fallen throughout the 2010s (Stelmach et al., 2021). A number of factors including pressures from the pandemic-induced recession, a rise in FIG transactions in the US, changes in accounting policies by the

ECB, and greater EU monetary and banking integration led to a post-crisis record value of transactions in 2020 (Jimenea and Lozabo, 2021). This has been further extenuated as industry sentiment has turned to growth and monetary conditions begin tightening (Jimenea and Lozano, 2021). In line with previous merger waves, Europe can be expected to follow consolidation activity among banks in America (Evanoff et al., 2009).

The aim of this paper is to assess whether bank M&A activity in Europe post-financial crisis created value for shareholders in the term immediately following the transactions, and if any variables significantly impacted how the market judged said transactions. The paper takes a quantitative approach to analyse deals using MacKinlay's event study in economics and finance framework (1997) to ascertain abnormal returns in acquirors' stock prices. These returns are then examined against a set of variables to determine any correlation arising with abnormal returns. The paper finds that a small majority (18 out of 30) of transactions created value for acquiring shareholders in the event window. It also finds a slightly negative correlation between abnormal returns and cross-border deals, and a slightly positive relationship between abnormal returns and deal size relative to acquirer size. The implication of these findings is discussed in the results section.

The paper aims to add to the contemporary literature on bank mergers and acquisitions by focusing particularly on transactions in the heavily bank-biased European continent (Langfield et al., 2016). Instead of considering American deals in the dataset, the US market is used as a vantage point for comparison. The paper also only uses post-financial crisis deals to avoid interpreting transactions in two fundamentally different economic environments together. Finally, it builds on the analysis of previous consolidation among European lenders in a time of significant restructuring for the sector - outside of intense merger activity – to understand how deal characteristics shape returns in a post-crisis financial environment. This has been further discussed in the literature review and results.

Literature Review

The wider European financial services industry has seen two recent periods of intense consolidation – merger waves (Evanoff et al., 2009). The first, beginning in the late nineties through till 1999 was a resultant factor of the creation of the common market area and the use of the common currency among many of the European Union's members. This was closely followed by another merger wave that coincided with the build-up of leverage in developed market economies from 2003 to 2007 (Lazarides, 2017). These merger waves closely followed similar periods of intense merger activity in the US market until 2006. Transactions following the 2008 financial crisis were largely driven by distressed sales like the acquisition of Lehman Brothers by Barclays (Hughes et al., 2008), or a wider restructuring of entire national financial sectors like those seen in Italy and Spain.

Merger waves often produce a mixed set of results on the financial performance of transactions. Since many of these transactions are done during a period of significant competition for targets, often a booming stock market, and relatively easy credit conditions, they may be seen as likely to destroy shareholder value (Alexandridis, 2011). However, this doesn't necessarily square with evidence from studies of consolidation among Europe's financial institutions during the noughties' bank mergers. Efficiency, gauged through revenue growth, profitability, and (or) cost reduction, has been shown to improve in the years following a merger in both European and American Banks (Hangendorff and Keasey, 2009) (Al-Sharkas et al., 2008). Enhancement in shareholder value, gauged using event studies to measure and attribute abnormal returns to the transaction, have also shown a net positive effect for European banks, albeit not for American banks (Scholtens and Wit, 2004). This may be attributed to both; the lower valuations in Europe compared to the US, and the fact that European banks may have the benefit of hindsight in learning from Americans' mistakes (Evanoff et al., 2009).

Pre-crisis European bank mergers were driven by a variety of different factors. While results have shown delivery of shareholder value and efficiency in performance, other factors including diversification and managerial hubris have been extensively studied as well. Evidence suggests that acquisition activity for European banks was driven by large acquirers purchasing large targets (Pasiouras et al., 2011). This research has also shown that more profitable banks with better growth prospects targeted less profitable banks with worse growth prospects. This may corroborate some broader arguments for M&A such as better management driving out less efficient managements and points to why share prices for acquirer tended to deliver positive abnormal returns.

In similar validation of existing M&A theories, non-diversifying mergers tended to create more value (Amihud et al., 2002). In spite of the cyclicality of banks in particular, there is very little evidence of any cashflow synergies or shareholder value creation achieved from product diversification into sectors with more stable earnings such as insurance (Ismail and Davidson, 2005). Geographic diversification though, has considerably more mixed returns. The use of a common currency and a single market for financial services in Europe created grounds for cross-border transactions. However, the lack of a European banking union has complicated the risk exposure pan-European banks face. This undermines the original intention among policymakers to allow European companies to enjoy the full scale of the European economy akin to their American counterparts (Calori & Lubatkin, 1994). In the fragmented European market, some research has suggested that focused domestic mergers create more value (Lensink and Maslennikova, 2008), while others have suggested the opposite (Ismail and Davidson, 2005).

Crisis era and post-crisis European banking consolidation has been studied through factorial changes in the financial services ecosystem including regulatory changes, changes in capital requirements, and distressed amalgamation of financial institutions. Bank M&A during the financial crisis were largely distressed transactions orchestrated by national governments to shore-up capital and liquidity in the financial system (Rao-Nicholson and Salaber, 2015). Research into wider results for shareholders in such deals is limited. This may be due to capital loss occurring regardless of the rescues of banks as the financial contagion spread across the industry. Shareholders were also often wiped out by rescuing governments taking positions in banks – making a comparable study hard to achieve.

Post-crisis sale of bank stakes by respective rescuing governments also has mixed results. The US government reported a net gain on the rescue while the British government made a profit on the sale of Lloyds but is losing money on as it returns NatWest to the market (Benink and Huizinga, 2013). It has been argued that a US-style rescue where the British government would have taken positions in all large lenders regardless of the need for capital infusion may have helped the government deliver a net gain. However, it is hard to nail these gains as a result of the restructuring and mergers during the crisis and not as a function of other factors like the macroeconomic recovery or asset purchases. Blanco's (2021) paper found that banking reforms and mergers in Spain during the crisis helped improve bank efficiency and solvency and allowed unviable banks to exit the market. However, a longer-term study of Italy's cooperative banks suggests that no significant cost efficiencies could be achieved until banks had reached considerable scale (Coccorese and Giovanni, 2019) – casting doubt upon the policy of merging banks to improve their positions in time of crisis.

Post-financial crisis, transactions in Europe continued to be driven as part of bank rescues. The causal link had shifted from the collapse of the US sub-prime mortgage market to the Eurozone debt crisis (Lazarides, 2017). In the years following the Eurozone debt crisis, a changing regulatory landscape became a key factor in mergers. In spite of a disinclination to allow more lenders to become systemically important (too-big-to-fail) in the early years after the financial crisis, greater tier 1 equity requirements actually worked as incentive for consolidation activity (Maragopoulos, 2021). Albeit consolidation was restricted among the largest banks in Europe (Schoenmaker and Duijim, 2022), Marques-Ibanez and Altunbas', (2004) analysis has shown that large banks have incentive to grow by acquiring smaller lenders. Some literature has evidenced that loan quality is major hindrance in bank M&A (Ismail et al., 2009). However, the tidying of bank balance sheets post-crisis created more incentive for consolidation. Changing rules have made banks fully liable for their subsidiaries which created further grounds for their full purchase and integration into the parent bank. Finally, overcapacity in the sector has been driving most of the current activity. In additional to the fundamental and regulatory changes in the industry discussed in previous sections, competition from non-bank lending entities (Allen and Gu, 2020) and a shift towards capital markets lending (Schoenmaker and Duijim, 2018) represent further structural changes for European banks.

The analysis of past literature suggests that while post-crisis European bank M&A has been studied extensively around the intention for mergers, there is little literature devoted to study of shareholder wealth creation during these transactions. The papers dedicated to studying shareholder wealth effects have either exclusively used pre-crisis datasets or bundled post-crisis transactions with other datasets such as US bank transactions. There could be several reasons for this. Through the 2010s, there was a wider decline in European banks' market capitalisations and profitability (Grodzicki et al., 2019) which could potentially distort results. There was also no intense consolidation period post-crisis to study similar to the late 90s and mid-noughties. Finally, the business environment for European banks was so quickly changing that it made like-for-like comparison relatively hard. These issues are later addressed in this section.

There was a general framework established by the study of shareholder wealth creation during the past consolidation periods. This can be majorly broken down into event studies that ascertain market reaction to the transactions and longer-term studies assessing improvements in operational performance of lenders following an acquisition. As previous analysis has shown, most papers assessed these figures for both targets and acquirers, and then considered returns along key factors such as the method of payment for the deal, domestic or cross-border mergers, size of acquirers and product diversifying deals. This paper will aim to use a similar framework in defining its hypothesis and eventual assessment.

Stárová, Teplý and Černohorský's (2010) paper conducts an event study to assess value creation of European bank M&A across 59 transactions from 1998 to 2007. It finds that while there was net value creation from the transactions, they destroyed value for bidders whilst transferring it to targets. They also establish that smaller transactions are better than larger transactions and that cash is a better alternative for payment compared to paper. However, they fail to find any difference between cross border and domestic transactions in terms of value creation in their dataset. There is radically different to Lensink and Maselmikona's (2007) paper which assess 75 deals in 19 countries from 1996 to 2004. They find positive and statistically significant abnormal gains for acquirers in contrast to the 2010 paper. They also argue that cross border transactions in fact couldn't create gains and in their domestic heavy-dataset, domestic deals created better value. They further go on to point out that organic growth abroad delivered better abnormal returns, which may be because bidding banks often fail to choose good international targets.

Ismail and Davidson's (2007) paper on similar analysis of shareholder wealth in 102 deals among European banks from 1987 to 1999 considers some additional factors. While they find that acquiring bank shares failed to live with significant abnormal returns either positive or negative, and targets delivered positive returns, they break this down into further constituents. They argue that cross border transactions in fact delivered better value compared to national deals which whilst gave positive abnormal returns, were still worse off than cross border transactions. However, cross-product transactions are found to be value destroying and focused bank-to-bank deals to created positive value. A mixed use of cash and equity delivered the best returns, the use of cash was next and finally equity only transactions delivered worst value. Incidentally they also find that European bidders seemed to be more cautious than American bidders which is represented by lower target returns among European banking transactions. They say this may be due to an expectation of falling profitability among European lenders and overcapacity in the sector.

Scholtens and Mit's (2004) paper studying 17 European targets and 20 bidders from 1990 to 1999 clearly establishes that there are clear announcement effects in bank mergers, and that these differ in Europe and the US. We find that in Europe both targets and bidders deliver positive abnormal returns whereas this is only true for targets in the US. This has also been corroborated by much of the existing literature before and after 2004. Moreover, this paper does not find significant difference between the returns from targets and bidders in the case of Europe. Cumulatively, the above four papers provide much of the framework for structuring this paper.

As for operational analysis and its relationship with shareholder value, Ismail, Davidson, and Frank's (2009) paper analysing 35 European banks from 1992 to 1997 found a decrease in profitability in combined entities. However, this comes with a corresponding improvement in cost-efficiency ratios, albeit not enough to offset the decrease in profitability. In their dataset, they find that, assuming an efficient market response (Fama, 1970), the market overestimated synergies to be drawn from transactions. While event studies deliver positive value, this didn't necessarily square with the future operational outcomes. They also add that a conservative credit policy and cost efficiency in targets pre-merger were the best indicators of enhanced operational returns. Altunbas and Ibáñez's (2004)

European Central Bank working paper agrees by showing that improved cost-efficiency was the primary post-merger strategy among European banks for both domestic and cross-border mergers. They also found that return on capital improved post-merger which could be driven by the loss of capitalisation in merged entities that the 2009 paper establishes. However, that would theoretically increase risk of bidder default, but this notion has been rejected (Vallascas and Hagendorff, 2011).

Given the above analysis, the issues around long-term loss in market capitalisation and changing business environment have been addressed through analysing only the short-term shareholder value creation for bidders instead of long-term operational studies. The concern around a small sample of deals can be rejected as the overall number of deals in the market are limited and contemporary literature has constantly relied on a small number of transactions to draw conclusions.

There are four hypotheses:

1. Acquiring banks deliver a positive abnormal return to shareholders
2. Cash acquisitions deliver higher abnormal returns compared to equity acquisitions
3. Domestic acquisitions delivered higher abnormal returns compared to cross-border acquisitions
4. Smaller acquisitions relative to bank size deliver higher abnormal returns compared to larger acquisitions relative to bank size

Data and Methodology

Mergers and Acquisitions have been studied widely using several different quantitative and qualitative means across industries for many varied stakeholders with different outcomes. Meglio and Risberg (2011) argue that there is no universal construct to measure M&A performance. Research has varied for the purposes of value creation, synergies realised, management outcomes, antitrust and other business metrics such as operational performance and investment. The paper also establishes that M&A performance has been gauged differently for different industries and geographies and that there are many differences in the long-term studies versus short-term assessments of value creation from M&A deals.

The financial industry had been in continuous reorganisation process through the 2010s and other external factors such as the Eurozone debt crisis affected banks' stock prices and accounting performance. This means there is limited comparable data to be used throughout this time period. As such, a short-term analysis of shareholder value creation has been deemed more fit for purpose. This paper takes a quantitative approach to analyse value creation for acquiring shareholders in the event window of a transaction. This has been the most widely used mechanism for analysis of M&A transactions throughout the literature (Sethi and Krishnakumar, 2010).

MacKinlay (1993) lays out the procedure for calculating cumulative abnormal returns delivered by a company's stock prices during an event window. This builds on the work of Brown and Warner (1985) who argue for a case in using daily stock prices in event studies. In effect, MacKinlay's (1993) approach requires the calculation of normal returns for a stock during the event window and using their difference with the actual returns to estimate abnormal returns over this period. The calculation of normal returns itself is given using an estimation window prior to the event on an undisturbed stock price with the following equation:

$$E(R_i) = \alpha_i + \beta_i M + \varepsilon$$

$$\text{Where, } \varepsilon = 0$$

This equation determines the expected returns for the stock in the absence of a transaction. M represents the returns of the benchmark over the event period. The error term is assumed to be zero. The alpha and beta are calculated by determining the constant and coefficient, respectively, of a regression of the stock price against the benchmark over the estimation period. MacKinlay (1993) determines the above market model using the CAPM model (Sharpe, 1964) and the Fama and French (1993) three-factor model. The use of regression means that its ability in giving accurate expected returns relies upon how well the variance for the stock price (dependant variable) is explained by the benchmark index (independent variable). Appendix A shows a high coefficient of determination, implying that the changes in the independent variable can explain the changes dependant to a very large degree. The market model used to predict returns is also sensitive to changes in alpha and beta. These have been computed to avoid using the figures that may not accurately represent the position of the stock at the point of the deal. It should also be noted that CAR will significantly depending on the market model. Appendix E shows how the use average stock returns over a period, for instance, changes the value of CAR.

Sethi and Krishnakumar (2010) explain that estimation windows have significantly differed in different papers and make a similar argument for transaction periods. Similar works such the Elferink (2021) paper have used estimation periods of 150 trading days - starting 20 trading days before the transaction: and event periods of 5 days - both before and after the transaction. Elferink (2021) argues that these choices are relatively arbitrary and there is a lack firm consensus in the literature about using any particular frameworks. This paper assumes an estimation window of 100 trading days, 20 trading days prior to the transaction announcement date. The choice of a shorter estimation window has been made to avoid encountering massive swings in prices due to external events such as the Eurozone Debt Crisis (Sethi and Krishnakumar, 2010). The event window is kept constant with the literature at 5 days each before and after the announcement.

Calculation of Alpha and Beta over the estimation window: T_{-120} to T_{-20}

Y variable – Acquiring Bank's Stock Price Daily Returns

X variable – Index Daily Returns (Euro STOXX 600 Banks)

Regression Intercept – Alpha

Regression Slope – Beta

Calculation of Abnormal return over the event window: T_{-5} to T_{+5}

$$AR_i = R_i - E(R_i)$$

$$CAR = Sum\ (Ar_i)$$

The computed value of CAR has been substituted to illustrate M&A performance. This value is reflected as a dependant variable that can be explained using some key deal characteristics. Das and Kapil (2012) paper list a wide range of independent variables that have been used to explain financial performance of Mergers and Acquisitions. The most widely available market measures from this list including deal size, acquirer size, method of payment (cash or equity), and deal's cross-border status have been considered – in line with the hypotheses. This paper has avoided the use of variables that require judgement decisions such as management experience in prior deals, or those that may require the computation of data outside the public domain like loan quality. The need for differentiating between product diversification and focused transactions has been avoided by only including bank targets and acquirers in the data.

$$CAR \Rightarrow \beta_0 + \beta_1 MoP + \beta_2 CB + \beta_3 Deal + \beta_4 Acquirer + \beta_5 (Deal/Acquirer)$$

The four explanatory variables used in the equation have been deemed significant in analysis of European banking transactions by much of the literature. Ismail and Davidson’s (2007) analysis of a set of EU bank deals reveals that cross-border and cash deals create more value than national and equity deals respectively. Azofra, Olalla, and Olmo’s (2008) paper found transaction size as an important factor in M&A in a previous merger wave. Further literature has pointed to the importance of factors such as deal relative to acquirer size (Teplý et al., 2010), the importance of efficiency in financial services (Ismail and Davidson, 2007), geographic diversification (Schoenmaker and Duijim, 2018), and product diversification (Ismail and Davidson, 2005). The variables finally inputted into the model have been determined by the publicly available information about the set of transactions and the hypotheses formed from the literature review.

To understand the importance of variables in the financial performance of deals given by CAR, the model is coded as a multiple linear regression. The full tabulated set of results is run through the code to determine the correlation of the explanatory variables with CAR. Owing to the small sample size, control variables about dates of transactions, or country of transaction are not deemed relevant. The output correlation matrix and scatter plots are then analysed to determine which variables are most relevant in explaining CAR. Since there is no future data for assessment, there is no need for a regression equation or a hypothesis test to support the relationship. Theoretically, the equation can be based on some part of the dataset and tested on the rest, but the small data size limits the ability of the code to accurately perform this.

The set of transactions used have been obtained from the Refinitiv terminal for publicly listed European acquiring banks from 1/1/11 to 31/12/20. All transactions in this set have been used in the statistical analysis to prevent any bias in results. However, transactions from the above set without publicly available data on stock prices have been excluded. Similarly, government or central bank-led recapitalisation of banks (bailouts), acquisition of European banks by non-European institutions, and private equity activities have been filtered out in the Refinitiv classification in favour of strategic mergers and acquisitions. The stock prices for the dates of the transaction and estimation periods have been sourced from S&P Capital IQ. Index values and market capitalisation (acquirer size) of the acquirer on the day of the deal announcement have also been sourced from S&P Capital IQ.

Since these transactions take place over a period of 10 years with significant changes in constituents of benchmark indices, the Euro STOXX 600 banks index has been uniformly used throughout this analysis as a benchmark in regressions. Data on announcement dates, methods of payment, cross-border status, and size of deals comes from the output set of deals from Refinitiv. All deal sizes are recorded in billions of dollars to avoid discrepancies caused by currencies values. Acquirer sizes have been recorded in billions of Euros owing to the listing of companies in Europe. The consistency, however, prevents arithmetical errors in the computed proportion of deal relative to acquirer size.

Results and Discussion

Appendix C represents the results of the computed cumulative abnormal returns for the acquiring banks in the transactions. The CAR values have been computed separately and inputted into the tabulated results (Appendix B). The method of payment has been analysed as a binary variable where 0 refers to equity and 1 refers to cash. Particular transactions have been specified as 0* where the deal has been paid for using a majority of equity and a small cash component. Owing to the complex structures of these transactions, these deals have been analysed as equity payments. Similarly, status of cross-border transactions is a binary variable where 1 is a cross-border acquisition and 0 is not. The country of the acquiring bank and year of transaction are part of the observations but haven't been used as control variables due to the small sample. Finally, the proportion of deal size to acquirer size has been computed from the adjoining data.

Overall, a small majority of transactions (18 out of 30, 60%) create value for shareholders. However, the overall sum and average of CAR is -24.2320% and -0.8077% respectively – implying value destruction for acquiring shareholders. Since the Eurobank Ergasias SA transaction is a clear outlier with a CAR of -102.668% (Table 2), the aggregates vary significantly without it. Ignoring the outlier, the sum and average of CAR is 78.4360% and 2.7047% respectively (Appendix C) - implying value creation for shareholders from the acquisitions. The value of CAR ranges from -11.516% to 22.792% and typically ranges between -1.5% to 3.5% with a slight skew to the right but generally normal distribution (Table 1). This seems to be the case regardless of acquirer size (Table 3), deal size (Table 4), or the proportion of deal to acquirer size (Table 5) as the graphs exhibit. The outlier largely remains to be Greece's Eurobank Ergasias SA.

Spain's CaixaBank SA has been the most active and largest dealmaker in the market (Table 6). This is followed by a distant Intesa Sanpaolo SPA and BBVA. Most of the largest deals are dominated by southern European nations – Spain, Italy, and Greece – due to the greater degree of restructuring in these markets following the financial and Eurozone crises. The largest acquirers outside of Southern Europe are the UK's CYBG PLC (now, Virgin Money) and France's BNP Paribas. Much of Western and Northern Europe saw supressed bank M&A through the 2010s owing to their relatively stronger post-crises economies and a lower need for distressed M&A. This has been blamed for growing overcapacity in the European banking sector (The Economist, 2019). Lower valuations for banks in Southern Europe also explain greater dealmaking in the region (The Economist, 2020). Greece suffers from the most value destruction for shareholders in the dataset – primarily driven by the losses from Eurobank. The sum of Spanish transactions has also been value destroying. Italy, on the other hand, has the most value creating sum of transactions, and most countries (7 out of 12) see net value creation from bank M&A (Table 7).

The consideration of acquiring banks relative to the proportion to deal to acquirer size reveals massively inflated figures for Greece's Eurobank Ergasias and Alpha Bank (Table 8). These extreme figures may be explained by the low market capitalisations of said bank due to the timing of transaction following the Eurozone debt crisis. In the case of Eurobank, the transaction came alongside a crash in the stock price (albeit macroeconomic conditions make it hard to isolate this on the deal). Alpha Bank, on the other hand, delivered positive abnormal returns (Appendix C). Outside of crisis era deals, UK's CYBG PLC had the largest transaction relative to its size. This is followed by Spain's Unicaja Banco, and Denmark's Ringkjobing and Vestjysk Banks (Table 6). Most of the large transactions created value for shareholders, but results remain relatively mixed (Appendix D) (Table 15).

$30 billion in transactions; over 80% of the deal value were domestic mergers with just under $7 billion in deals being classified as cross-border transactions (Table 9). This diverges from the number of cross-border deals which comes in just under one-third (9 out of 30). A clear inference is that cross-border acquisitions have been smaller than average in size and fewer in count among Europe's publicly listed banks from 2011 to 2020. Four out of the nine cross-border transactions created value for shareholders whilst other destroyed value. The average CAR for cross-border deals was -1.03%, compared to an average CAR of 4.18% for domestic mergers excluding Eurobank, and -0.71% including Eurobank (Table 12). CaixaBank and Credit Agricole created the most value from cross-border mergers (Table 13). CaixaBank stands out because nearly 67% of the value created across its 5 transactions came from cross-border mergers (Table 11). This is also particularly peculiar because the geographic distribution also diverges from the norm with greater cross-border deal value for French and Eastern European banks compared to the crisis-hit southern European banks (Table 10). This may be explained both by the size of southern European banks (The Economist, 2020) and confidence in their respective economies.

46% of deal value has been paid for using cash with the remaining 54% paid using equity or both cash and equity (Table 14). This differs from the count as only 13 deals are paid for using equity or equity and cash, and 17 are cash-only transactions. This implies a higher-than-average size of equity deals and vice versa. Both equity and cash transactions seem to create value for shareholders. An equity deal had an average CAR 2.53%, and a cash deal had an average CAR of 2.73% excluding Eurobank or -5.17% with Eurobank (Table 15). This conforms with the belief that cash deals create more value than equity transactions (Evanoff et al., 2009). 5 of the 13 equity deals seem are concentrated during the Eurozone debt crisis – mostly in Spain. This may suggest that banks prefer using cash in transaction to prevent dilution of their shareholders in normal times. Choice of payment method may also have been affected by the kind of transaction involved – reorganisation of business and full integration of partly-owned subsidiaries has been a repeated feature in this dataset. Such deals are mostly paid for using cash.

The second aim of this paper is to understand the relationship between the financial performance of the transactions given by abnormal returns and the independent variables. The output from the multiple linear regression analysis coded for the above set of results has yielded a correlation matrix (Table 16) (Table 17). The correlation matrix shows that CAR is not highly correlated to any of the given variables. It is shown to be moderately positively correlated with deal size, acquirer size, and cross-border status, and moderately negatively correlated to the proportion of deal size to acquirer size. CAR shows very slight correlation with the method of payment. The independent variables' relationships with each other are rather limited as well except a moderate correlation for cross-border status with acquirer size and method of payment.

The inverse relationship between CAR and the proportion deal size to acquirer size is the most significant. This effectively implies that smaller deals as a proportion of a bank's market capitalisation are likely to create greater value. The relationship shows a higher relevance to CAR than both its individual constituents – deal size and acquirer size. However, the relationship may have been highly influenced by the sharp losses to Eurobank's stock after its acquisitions and be skewed by this outlier. This seems especially possible given that the next largest deals as a percentage of market capitalisation including CYBG, Ringkjobing and Vestjysk – all delivered positive returns. The large size of the losses on Eurobank seems to supress their effects. Finally, there is a slight correlation of CAR with the method of payment. This may have also been influenced by the outlier in the data set. The correlation is likely to be even lower as evidenced by the arithmetic from the previous analysis (Table 15). The method of payment – either in cash or paper – didn't radically alter CAR from the deals.

In a correlation study excluding the outlier, results change considerably (Appendix D). A slight relationship is established between value destruction and cross-border transactions. There is no relationship found between CAR and method pf payment, and the inverse relationship between CAR and the proportion of deal and acquirer size changes to a slightly positive correlation. These results in effect negate much of the correlation studied with the outlier. Across the five metrics used to explain the CAR, none of the given variables seem to explain the financial performance of the deals in a greatly significant manner. The explanatory variables are themselves not highly correlated to each other. This can be viewed positively as meaningfully different types of variables were used in trying to explain the abnormal returns. A final regression is deemed unnecessary, both due to the lack of future data for hypothesis testing and the lack of significant correlations established between dependant and independent variables.

With respect to the original hypotheses posed in this paper, the first and third hypotheses have been accepted and the second and fourth have been rejected. The tabulated results show that a majority of transactions created value for the acquiring shareholders, and this was also true on average. The analysis of the deals individually and the correlation matrix both suggest that cross-border transactions were value-destroying for shareholders on average. It also finds significantly larger positive abnormal returns from domestic transactions. The abnormal returns from both cash and equity deals were relatively close in size. The correlation matrix confirms that there was no relationship between returns and method of payment in this dataset. Finally, the relationship between CAR and the proportion of deal to acquirer size was slightly positive, ignoring the outlier. This suggests that larger deal sizes as a proportion of acquirer size created more value and vice versa.

In context of the literature, positive abnormal returns for acquiring banks are in line with prior research. While some papers (Teplý et al., 2010) have cast doubt upon this assumption, much of literature has found positive abnormal returns for acquiring banks' shares in Europe. The literature has been relatively divided on the ability of cross-border transactions to create value. This paper attests to the destruction of shareholder wealth from cross-border deals on average and a relationship between negative abnormal returns and cross-border deals. While the Teplý et al. (2010) paper ascertained that smaller deals are more likely to deliver shareholder value, the analysis on the given dataset fails to corroborate this notion. In fact, there has been a slight positive relationship between larger deals sizes as a proportion of the acquirer size found in this paper. Finally, this paper also rejects the hypothesis that cash deals are more likely to create value for shareholders – in contrast with much of the literature (Evanoff et al., 2009). This is however, also explained using the different datasets in use and the different economic environment in which these deals were proposed.

The results from this analysis may contain important implication in assessing post-crisis era bank deals in Europe. This paper uses pre-existing frameworks to derive meaningful results which contains both; the latest transaction in European bank deals, and a dataset more relevant to the post-financial crisis banking sector. Unlike an assessment of bull market merger waves and transaction in past papers, this paper's novelty lies in its assessment of value creation in an industry during a period of significant restructuring and distress. The paper also considers the changed nature of motivations for transactions including regulations, integration of subsidiaries and capital preservation, and posits that may these have driven investor reaction to said deals. The results contain implications in understanding how investors reward transactions during a period of crisis and reorganisation. The lack of any significant premium in abnormal returns on cash deals in an important example of this. There are also implications for European regulators and policymakers that are seeking to drive pan-European bank consolidation (Calori & Lubatkin, 1994) with results found for cross-border deals. Finally, the paper extends on the prior literature's findings by demonstrating that European bank M&A have continued to deliver value for acquiring shareholders in the post-financial crisis period.

Conclusion and Limitations

This paper studies the shareholder value effect of European bank mergers and acquisitions in the post-crisis period. It establishes that the European banking sector has undergone a long period of reorganisation following the financial and Eurozone debt crises. In this period, the underlying fundamentals of industry have changed and caused overcapacity in the industry which can be reduced by amalgamating financial institutions. The aims of consolidation activity have changed post-crisis to include capitalisation concerns, regulatory changes, competitive threats, and reducing loss of capital.

The paper evaluates transactions over the 2010s to assess for creation of shareholder value. It uses the framework practiced in assessment of prior merger waves and an event study analysis over the dataset of European public market deals. There also a consideration of evaluating these returns as an outcome of transaction-specific factors including size, geography, and method of payment. The study finds that 60% of transactions created value for shareholders but fails to establish any significant relationship between the chosen factors and abnormal returns. It also specifies that differences with and without the presence of outlier in the aggregates, averages, and correlations.

The paper accepts the hypotheses that deals created value for shareholders and that domestic transactions create greater value than cross-border transactions. It rejects the hypotheses that cash deals create greater value than equity deals and, that smaller deals create greater value than larger deals. The inability to draw any relationship between method of payment and returns also differs from the literature, albeit the aggregates conform to it. The findings imply some divergence between post-crisis and pre-crisis results which carry important implications for deal structures, understanding investor sentiment, and for policymakers. However, results need to be considered in full context of the macroeconomic environment of Europe and rely upon the dataset of given deals.

The use of the Euro STOXX 600 Banks index as a benchmark across the dataset acts as a weakness in some statistical calculations. Due to significant upheaval in national indices post-crisis, and to ensure uniformity, the given benchmark was chosen. While most acquiring banks were constituents of index at the time, their respective returns diverged greatly during times of external shocks, undermining calculation of normal returns. The paper is also limited in its ability to establish significant relationships between CAR and the explanatory variables. While the interpretation of the results provides some insight into which variables affect or don’t affect the abnormal returns, there are no significant correlations found. This can be the subject of further research by considering the ability of different explanatory variables such as loan quality and management experience in determining CAR. Similarly, the lack of comparable deals hurdles the process of validating the results of this paper. Further studies of bank deals during periods of distress in other geographies and of other industries may act as proxies to validate these results.

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Appendices and Tables

Appendix A: Example for Calculation for Constant and Coefficient of Regression

Dates	Crédit Agricole	Change	Date	Euro STOXX	Change
Jun-01-2020	8.02		Jun-01-2020	91.62	
Jun-02-2020	8.20	0.022954092	Jun-02-2020	94.24	0.0285964
Jun-03-2020	8.48	0.034634146	Jun-03-2020	98.16	0.0415959
Jun-04-2020	8.55	0.008250825	Jun-04-2020	98.07	-0.000917
Jun-05-2020	9.10	0.06336217	Jun-05-2020	104.14	0.0618946
Jun-08-2020	9.28	0.020228672	Jun-08-2020	105.78	0.015748
Jun-09-2020	8.69	-0.063362069	Jun-09-2020	102.01	-0.03564
Jun-10-2020	8.58	-0.013115509	Jun-10-2020	100.69	-0.01294
Jun-11-2020	8.06	-0.060853346	Jun-11-2020	93.73	-0.069123
Jun-12-2020	8.13	0.008689176	Jun-12-2020	94.35	0.0066147
Jun-15-2020	8.06	-0.008122077	Jun-15-2020	93.35	-0.010599
Jun-16-2020	8.34	0.034243176	Jun-16-2020	96.85	0.0374933
Jun-17-2020	8.30	-0.004798464	Jun-17-2020	96.40	-0.004646
Jun-18-2020	8.24	-0.007232401	Jun-18-2020	95.33	-0.0111
Jun-19-2020	8.07	-0.020641088	Jun-19-2020	95.02	-0.003252
Jun-22-2020	8.08	0.001983635	Jun-22-2020	94.09	-0.009787
Jun-23-2020	8.40	0.039346696	Jun-23-2020	96.56	0.0262515
Jun-24-2020	8.04	-0.043333333	Jun-24-2020	93.08	-0.03604
Jun-25-2020	8.28	0.030114485	Jun-25-2020	94.55	0.0157929
Jun-26-2020	8.08	-0.023435612	Jun-26-2020	92.46	-0.022105
Jun-29-2020	8.39	0.037357744	Jun-29-2020	94.59	0.023037
Jun-30-2020	8.43	0.005008347	Jun-30-2020	94.11	-0.005075
Jul-01-2020	8.38	-0.005932606	Jul-01-2020	93.49	-0.006588
Jul-02-2020	8.75	0.044640726	Jul-02-2020	97.48	0.0426784
Jul-03-2020	8.72	-0.003884826	Jul-03-2020	96.18	-0.013336
Jul-06-2020	8.96	0.027988071	Jul-06-2020	99.92	0.0388854
Jul-07-2020	8.89	-0.00848025	Jul-07-2020	98.52	-0.014011
Jul-08-2020	8.76	-0.014179608	Jul-08-2020	96.84	-0.017052
Jul-09-2020	8.56	-0.023287671	Jul-09-2020	94.81	-0.020962
Jul-10-2020	8.85	0.034595605	Jul-10-2020	97.08	0.0239426
Jul-13-2020	8.95	0.010619069	Jul-13-2020	98.39	0.013494
Jul-14-2020	9.01	0.007154035	Jul-14-2020	98.22	-0.001728
Jul-15-2020	9.13	0.013318535	Jul-15-2020	99.68	0.0148646
Jul-16-2020	9.22	0.009419496	Jul-16-2020	99.83	0.0015048
Jul-17-2020	9.11	-0.011067708	Jul-17-2020	98.87	-0.009616
Jul-20-2020	9.13	0.001755541	Jul-20-2020	99.17	0.0030343
Jul-21-2020	9.21	0.00898138	Jul-21-2020	100.13	0.0096803
Jul-22-2020	9.00	-0.023013461	Jul-22-2020	98.66	-0.014681
Jul-23-2020	8.99	-0.001555556	Jul-23-2020	97.60	-0.010744
Jul-24-2020	8.91	-0.008680169	Jul-24-2020	96.68	-0.009426
Jul-27-2020	8.72	-0.02065559	Jul-27-2020	94.57	-0.021825
Jul-28-2020	8.75	0.002521779	Jul-28-2020	94.98	0.0043354
Jul-29-2020	8.65	-0.010976446	Jul-29-2020	93.91	-0.011266
Jul-30-2020	8.24	-0.046936416	Jul-30-2020	89.91	-0.042594
Jul-31-2020	8.13	-0.01431344	Jul-31-2020	89.05	-0.009565
Aug-03-2020	8.37	0.030273197	Aug-03-2020	90.45	0.0157215
Aug-04-2020	8.54	0.019589107	Aug-04-2020	91.89	0.0159204
Aug-05-2020	8.53	-0.000234302	Aug-05-2020	91.89	0
Aug-06-2020	8.44	-0.010780408	Aug-06-2020	91.13	-0.008271
Aug-07-2020	8.45	0.001421464	Aug-07-2020	90.68	-0.004938
Aug-10-2020	8.61	0.018689378	Aug-10-2020	92.53	0.0204014
Aug-11-2020	9.03	0.048536925	Aug-11-2020	96.08	0.0383659
Aug-12-2020	9.09	0.006423034	Aug-12-2020	96.72	0.0066611
Aug-13-2020	8.91	-0.020026408	Aug-13-2020	94.87	-0.019127
Aug-14-2020	8.79	-0.013249495	Aug-14-2020	94.32	-0.005797
Aug-17-2020	8.69	-0.010923987	Aug-17-2020	93.19	-0.01198
Aug-18-2020	8.57	-0.014265992	Aug-18-2020	92.26	-0.00998
Aug-19-2020	8.70	0.015172736	Aug-19-2020	93.69	0.0154997
Aug-20-2020	8.46	-0.027132674	Aug-20-2020	91.56	-0.022735
Aug-21-2020	8.33	-0.015362798	Aug-21-2020	90.93	-0.006881
Aug-24-2020	8.57	0.029044647	Aug-24-2020	93.13	0.0241944
Aug-25-2020	8.60	0.002565897	Aug-25-2020	92.96	-0.001825
Aug-26-2020	8.67	0.008143322	Aug-26-2020	93.67	0.0076377
Aug-27-2020	8.62	-0.005769675	Aug-27-2020	92.95	-0.007687
Aug-28-2020	8.90	0.032497679	Aug-28-2020	94.53	0.0169984
Aug-31-2020	8.59	-0.034622302	Aug-31-2020	92.58	-0.020628
Sep-01-2020	8.51	-0.008616674	Sep-01-2020	91.15	-0.015446
Sep-02-2020	8.44	-0.009161381	Sep-02-2020	90.56	-0.006473
Sep-03-2020	8.44	0.000237079	Sep-03-2020	90.25	-0.003423
Sep-04-2020	8.67	0.027494667	Sep-04-2020	91.69	0.0159557
Sep-07-2020	8.74	0.008535179	Sep-07-2020	92.21	0.0056713
Sep-08-2020	8.52	-0.026075023	Sep-08-2020	90.15	-0.02234
Sep-09-2020	8.66	0.016674495	Sep-09-2020	91.52	0.0151969
Sep-10-2020	8.69	0.003234003	Sep-10-2020	91.09	-0.004698
Sep-11-2020	8.52	-0.018650702	Sep-11-2020	89.60	-0.016357
Sep-14-2020	8.64	0.013843266	Sep-14-2020	90.31	0.0079241
Sep-15-2020	8.54	-0.012034251	Sep-15-2020	89.31	-0.011073
Sep-16-2020	8.45	-0.01054111	Sep-16-2020	89.79	0.0053745
Sep-17-2020	8.36	-0.010416667	Sep-17-2020	88.33	-0.01626
Sep-18-2020	8.10	-0.031578947	Sep-18-2020	86.00	-0.026378
Sep-21-2020	7.66	-0.053606719	Sep-21-2020	81.10	-0.056977
Sep-22-2020	7.62	-0.005742626	Sep-22-2020	80.93	-0.002096
Sep-23-2020	7.56	-0.007613547	Sep-23-2020	80.57	-0.004448
Sep-24-2020	7.42	-0.018253968	Sep-24-2020	80.26	-0.003848
Sep-25-2020	7.16	-0.03476152	Sep-25-2020	79.27	-0.012335
Sep-28-2020	7.58	0.057788945	Sep-28-2020	83.73	0.0562634
Sep-29-2020	7.40	-0.023225125	Sep-29-2020	81.87	-0.022214
Sep-30-2020	7.47	0.009456904	Sep-30-2020	82.68	0.0098937
Oct-01-2020	7.34	-0.017933619	Oct-01-2020	82.42	-0.003145
Oct-02-2020	7.25	-0.011719815	Oct-02-2020	83.06	0.0077651
Oct-05-2020	7.38	0.018201875	Oct-05-2020	84.27	0.0145678
Oct-06-2020	7.77	0.05227519	Oct-06-2020	87.11	0.0337012
Oct-07-2020	7.67	-0.013384813	Oct-07-2020	86.65	-0.005281
Oct-08-2020	7.78	0.014609966	Oct-08-2020	87.80	0.0132718
Oct-09-2020	7.67	-0.013885318	Oct-09-2020	87.14	-0.007517
Oct-12-2020	7.71	0.005215124	Oct-12-2020	87.00	-0.001607
Oct-13-2020	7.42	-0.037872892	Oct-13-2020	84.62	-0.027356
Oct-14-2020	7.49	0.00970612	Oct-14-2020	84.59	-0.000355
Oct-15-2020	7.25	-0.032309746	Oct-15-2020	83.31	-0.015132
Oct-16-2020	7.34	0.013245033	Oct-16-2020	84.79	0.017765
Oct-19-2020	7.39	0.005718954	Oct-19-2020	85.55	0.0089633
Oct-20-2020	7.44	0.006769564	Oct-20-2020	86.13	0.0067797
Oct-21-2020	7.32	-0.015061861	Oct-21-2020	85.08	-0.012191
Oct-22-2020	7.38	0.00737302	Oct-22-2020	85.55	0.0055242
Oct-23-2020	7.52	0.018704256	Oct-23-2020	87.96	0.0281707
	Alpha	-4.31562E-05			
	Beta	1.097455789			

SUMMARY OUTPUT

Regression Statistics	
Multiple R	0.94255484
R Square	0.888409626
Adjusted R Square	0.887282451
Standard Error	0.008139493
Observations	101

ANOVA

	df	*SS*	*MS*	*F*	*Significance F*
Regression	1	0.052217542	0.052217542	788.1733003	6.10526E-49
Residual	99	0.006558883	6.62513E-05		
Total	100	0.058776425			

	Coefficients	*Standard Error*	*t Stat*	*P-value*	*Lower 95%*	*Upper 95%*	*Lower 95.0%*	*Upper 95.0%*
Intercept	-4.31562E-05	0.00081061	-0.053239127	0.957648664	-0.001651583	0.001565271	-0.001651583	0.001565271
X Variable 1	1.097455789	0.039090945	28.07442431	6.10526E-49	1.019890873	1.175020706	1.019890873	1.175020706

Appendix B: Example for Calculation of Normal and Cumulative Abnormal Returns (CAR)

Dates	Crédit Agricole S.A. (ENXTPA:ACA) - Share Pricing	Change	Expected	Abnormal	Dates	STOXX	Change
Nov-11-2020	8.76				Nov-11-2020	103.17	
Nov-12-2020	8.54	-0.024211969	-0.02121149	-0.003000479	Nov-12-2020	101.18	-0.019288553
Nov-13-2020	8.63	0.010065543	0.012755795	-0.002690251	Nov-13-2020	102.36	0.011662384
Nov-16-2020	8.79	0.019003476	0.03651726	-0.017513784	Nov-16-2020	105.77	0.033313794
Nov-17-2020	8.82	0.003183989	-0.000769467	0.003953456	Nov-17-2020	105.70	-0.000661813
Nov-18-2020	8.92	0.011562004	0.013454408	-0.001892404	Nov-18-2020	107.00	0.012298959
Nov-19-2020	8.87	-0.005826983	-0.011940808	0.006113824	Nov-19-2020	105.84	-0.010841121
Nov-20-2020	8.95	0.008566276	0.002963856	0.00560242	Nov-20-2020	106.13	0.002739985
Nov-23-2020	9.31	0.040679481	0.017122364	0.023557118	Nov-23-2020	107.79	0.015641195
Nov-24-2020	9.82	0.054768041	0.047096841	0.0076712	Nov-24-2020	112.42	0.042953892
Nov-25-2020	9.87	0.005294237	-0.005509935	0.010804172	Nov-25-2020	111.86	-0.00498132
Nov-26-2020	9.83	-0.0046587	-0.012895535	0.008236835	Nov-26-2020	110.55	-0.011711067
Nov-27-2020	9.86	0.003663004	0.008593538	-0.004930534	Nov-27-2020	111.42	0.007869742
Nov-30-2020	9.67	-0.019261963	-0.030478791	0.011216828	Nov-30-2020	108.33	-0.027732903
			CAR	0.052819132			
			Alpha	-4.31562E-05			
			Beta	1.097455789			

Appendix C: Tabulated Results of CAR for Acquiring Banks

Year	Acquiring Bank	CAR	MoP	Country	Cross-Border	Acquirer Size	Deal Size	Deal Size	Deal Size/ Acquirer Size
2011	Banco Popular Espanol SA	-4.1970%	0	Spain	0	4.96	1.464861033	1,464.86	0.295334886
2012	Bank Zachodni WBK SA	0.1540%	0	Poland	0	3.9816	1.425464099	1,425.46	0.358012884
2012	CaixaBank SA	-11.5160%	0	Spain	0	11.37	1.30524263	1,305.24	0.114797065
2012	National Bank of Greece SA	13.7220%	0	Greece	0	2	0.708763424	708.76	0.354381712
2012	Banco Santander SA	3.4020%	0	Spain	0	59.93	0.346702031	346.70	0.005785116
2012	Alpha Bank	11.1210%	1	Greece	0	0.94566	2.84	2,840.00	3.003193537
2013	Powszechna Kasa Oszczednosci Bank Polski SA	5.7900%	1	Poland	0	10.9272	0.82583	825.83	0.075575628
2013	Eurobank Ergasias SA	-102.6680%	0	Greece	0	0.25639	0.88952	889.52	3.469402083
2013	BNP Paribas SA	-2.5290%	1	France	1	70.02	1.33013	1,330.13	0.01899643
2014	CaixaBank SA	4.0260%	1	Spain	0	25.66	2	2,000.00	0.077942323
2014	OTP Banka Hrvatska dd	-4.3170%	1	Croatia	1	3.841986011	0.14434	144.34	0.037569111
2014	Banco Di Desio E Della Brianza SpA	22.7920%	1	Italy	0	0.42491	0.19274	192.74	0.453601939
2014	BNP Paribas SA	3.5340%	1	France	1	61.55	0.47385	473.85	0.007698619
2014	BBVA	-1.4870%	1	Spain	0	53.76	3.25	3,250.00	0.060453869
2015	Banco de Sabadell SA	-10.5230%	1	Spain	1	9.1	2.52976	2,529.76	0.277995604
2016	Alior Bank SA	9.2140%	1	Poland	0	1.1952	0.36681	366.81	0.30690261
2016	CaixaBank SA	-1.3370%	1	Spain	1	15.35	0.71877	718.77	0.046825407
2018	Ringkjobing Landbobank A/S	8.4680%	0*	Denmark	0	1.0543	0.60412	604.12	0.573005786
2018	CaixaBank SA	6.7540%	1	Spain	1	25.09	0.12289	122.89	0.004897967
2018	CYBG PLC	1.4410%	0	United Kingdom	0	3.249	2.11471	2,114.71	0.650880271
2018	Basler Kantonalbank AG	4.7800%	1	Switzerland	0	2.6535	0.21446	214.46	0.080821556
2018	Erste Group Bank AG	1.6160%	1	Austria	1	14.26	0.09186	91.86	0.006441795
2019	BPER Banca SpA	15.9400%	0*	Italy	0	1.58	0.28289	282.89	0.179044304
2019	Commerzbank Inlandsbanken Holding GmbH	-3.9720%	1	Germany	0	6.69	0.35459	354.59	0.05300299
2020	Intesa Sanpaolo Spa	-1.8770%	0*	Italy	0	44.43	4.77295	4,772.95	0.107426289
2020	Nova Ljubljanska Banka dd Ljubljana	-7.7220%	1	Slovenia	1	1.24	0.42103	421.03	0.339540323
2020	CaixaBank SA	10.1540%	0	Spain	0	10.85	5.14009	5,140.09	0.473741014
2020	Unicaja Banco SA	-0.3310%	0	Spain	0	1.17	0.71609	716.09	0.612042735
2020	Credit Agricole Italia SpA	5.2820%	1	Italy	1	26.84	1.04751	1,047.51	0.039027943
2020	Vestjysk Bank A/S	0.0540%	0	Denmark	0	0.3588	0.19773	197.73	0.551086957
			* Cash + Equity			in billions of €	in billions of $		
	Avg without Eurobank	2.7047%							
	Avg with Eurobank	-0.8077%							
	Sum without Eurobank	78.4360%							
	Sum with Eurobank	-24.2320%							
	Minimum without Eurobank	-11.5160%							
	Minimum with Eurobank	-102.6680%							
	Maximum	22.7920%							

Appendix D: Correlation Matrix without outliers

	CAR	*MoP*	*Cross-Border*	*Acquirer Size*	*Deal Size*	*DAS*
CAR	1					
MoP	-0.027231692	1				
Cross-Border	-0.329178067	0.563601862	1			
Acquirer Size	-0.175731717	0.173970505	0.288377141	1		
Deal Size	-0.108104301	-0.222161541	-0.24285775	0.213449118	1	
DAS	0.269299951	-0.061711354	-0.281138287	-0.346875193	0.256628095	1

Appendix E: Example of CAR with Average Returns (Credit Agricole – Refer to Appendix A & B)

Dates	Crédit Agricole S.A. (ENXTPA:ACA) - Share Pricing	Change	Average	Abnormal	Dates	STOXX	Change
Nov-11-2020	8.76				Nov-11-2020	103.17	
Nov-12-2020	8.54	-0.024211969	-0.000861975	-0.023349994	Nov-12-2020	101.18	-0.019288553
Nov-13-2020	8.63	0.010065543	-0.000861975	0.010927518	Nov-13-2020	102.36	0.011662384
Nov-16-2020	8.79	0.019003476	-0.000861975	0.019865451	Nov-16-2020	105.77	0.033313794
Nov-17-2020	8.82	0.003183989	-0.000861975	0.004045964	Nov-17-2020	105.70	-0.000661813
Nov-18-2020	8.92	0.011562004	-0.000861975	0.012423979	Nov-18-2020	107.00	0.012298959
Nov-19-2020	8.87	-0.005826983	-0.000861975	-0.004965009	Nov-19-2020	105.84	-0.010841121
Nov-20-2020	8.95	0.008566276	-0.000861975	0.00942825	Nov-20-2020	106.13	0.002739985
Nov-23-2020	9.31	0.040679481	-0.000861975	0.041541456	Nov-23-2020	107.79	0.015641195
Nov-24-2020	9.82	0.054768041	-0.000861975	0.055630016	Nov-24-2020	112.42	0.042953892
Nov-25-2020	9.87	0.005294237	-0.000861975	0.006156212	Nov-25-2020	111.86	-0.00498132
Nov-26-2020	9.83	-0.0046587	-0.000861975	-0.003796725	Nov-26-2020	110.55	-0.011711067
Nov-27-2020	9.86	0.003663004	-0.000861975	0.004524978	Nov-27-2020	111.42	0.007869742
Nov-30-2020	9.67	-0.019261963	-0.000861975	-0.018399988	Nov-30-2020	108.33	-0.027732903
			CAR	12.65%			
			Alpha	-4.31562E-05			
			Beta	1.097455789			

Table 1

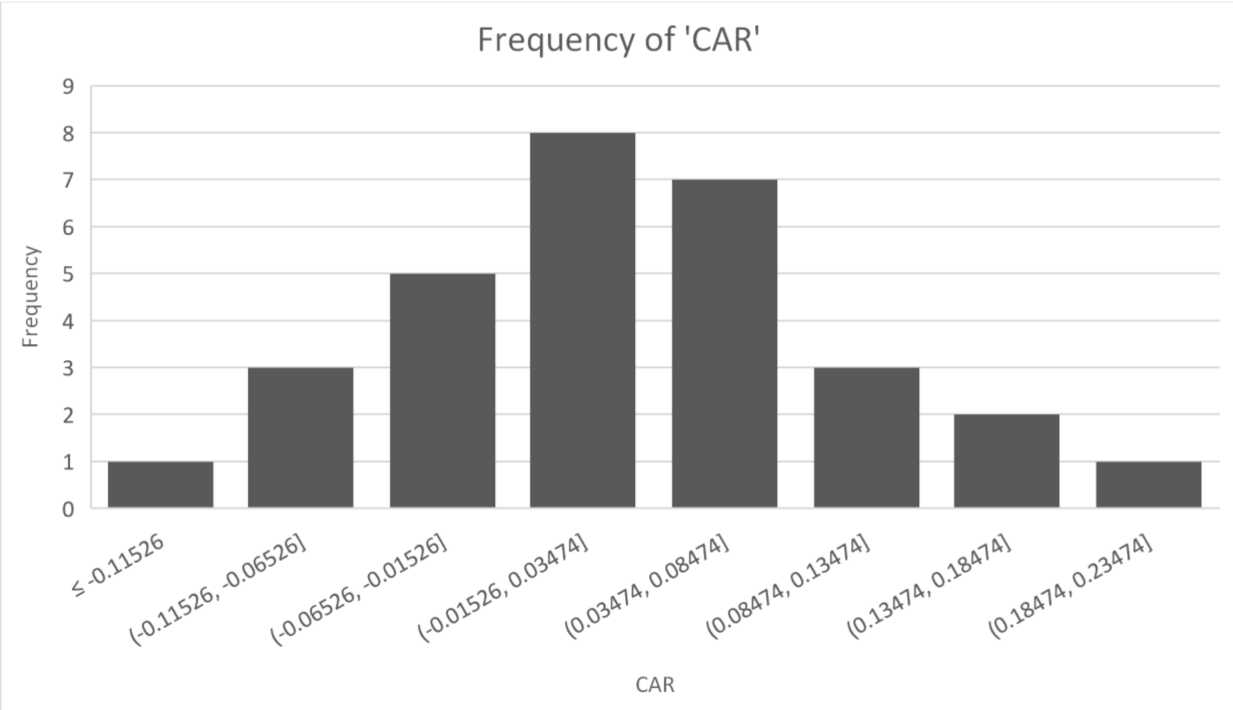

Frequency of 'CAR'
9
8
7
6
5
4
3
2
1
0
Frequency
≤ -0.11526
(-0.11526, -0.06526]
(-0.06526, -0.01526]
(-0.01526, 0.03474]
(0.03474, 0.08474]
(0.08474, 0.13474]
(0.13474, 0.18474]
(0.18474, 0.23474]
CAR

Table 2

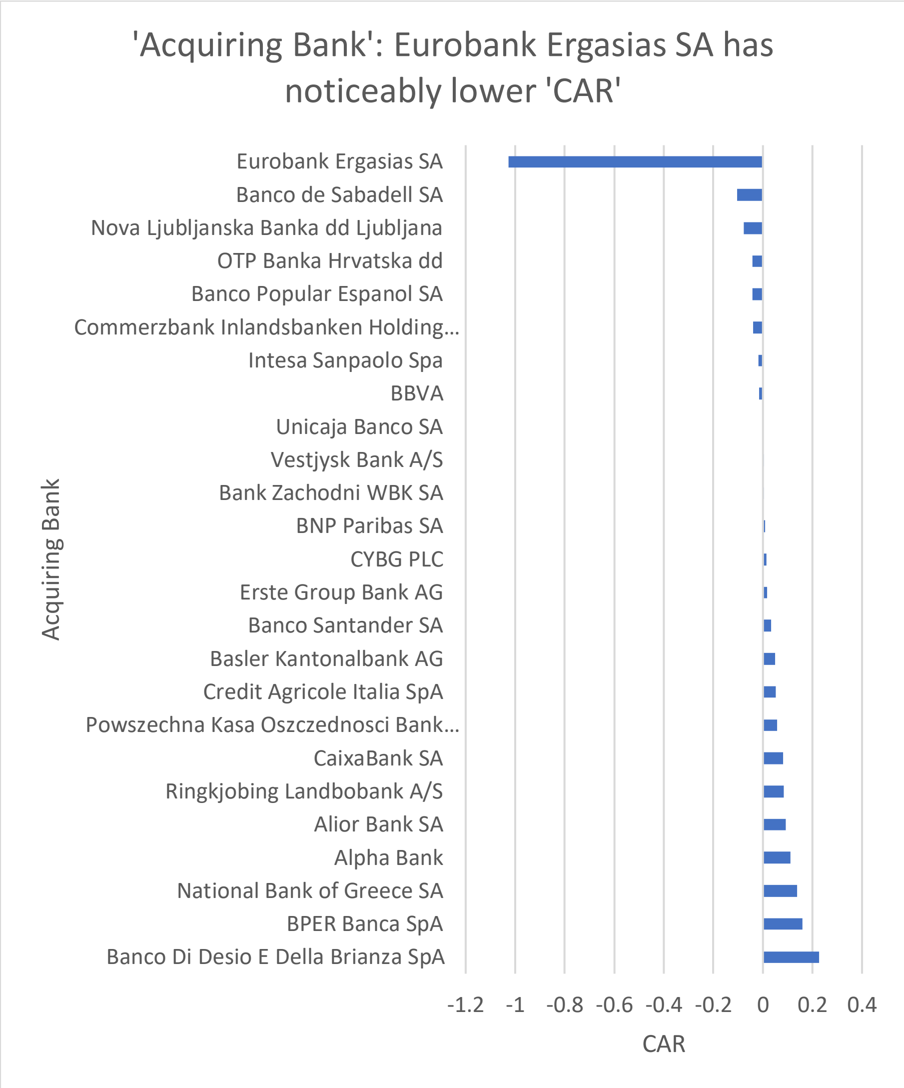

'Acquiring Bank': Eurobank Ergasias SA has noticeably lower 'CAR'
Acquiring Bank
Eurobank Ergasias SA
Banco de Sabadell SA
Nova Ljubljanska Banka dd Ljubljana
OTP Banka Hrvatska dd
Banco Popular Espanol SA
Commerzbank Inlandsbanken Holding...
Intesa Sanpaolo Spa
BBVA
Unicaja Banco SA
Vestjysk Bank A/S
Bank Zachodni WBK SA
BNP Paribas SA
CYBG PLC
Erste Group Bank AG
Banco Santander SA
Basler Kantonalbank AG
Credit Agricole Italia SpA
Powszechna Kasa Oszczednosci Bank...
CaixaBank SA
Ringkjobing Landbobank A/S
Alior Bank SA
Alpha Bank
National Bank of Greece SA
BPER Banca SpA
Banco Di Desio E Della Brianza SpA
-1.2
-1
-0.8
-0.6
-0.4
-0.2
0
0.2
0.4
CAR

Table 3

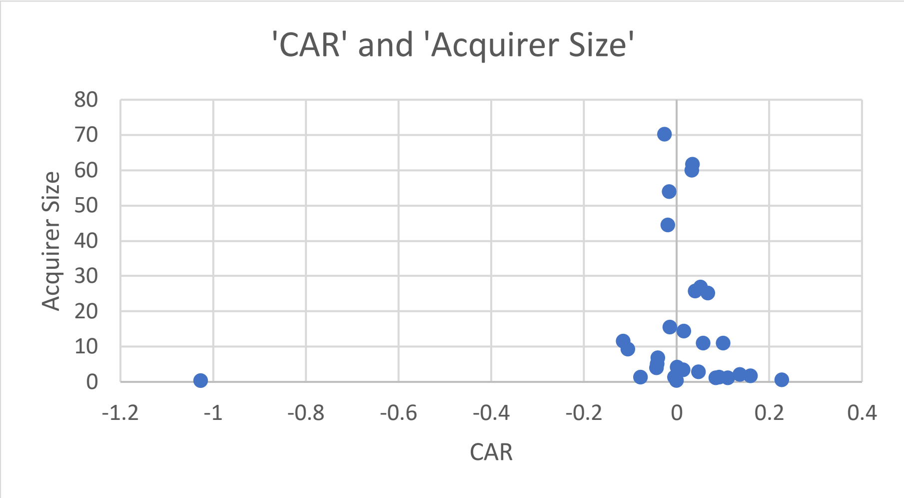


Table 4

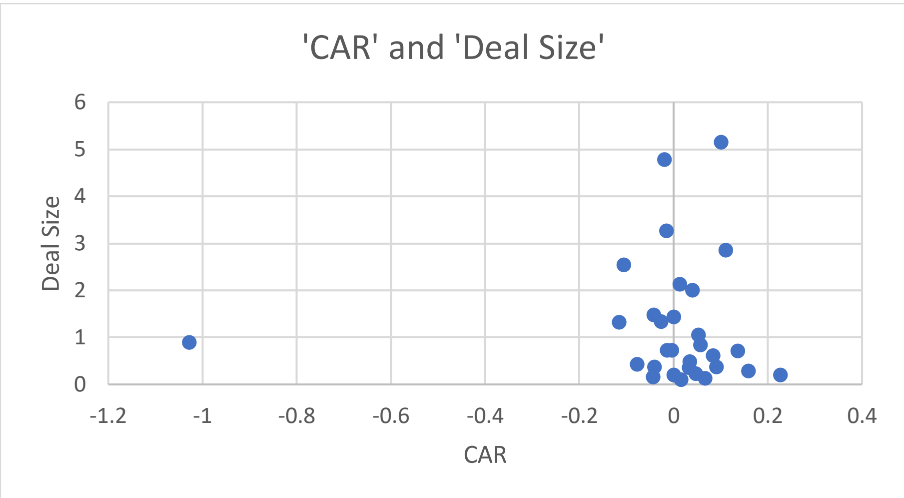


Table 5

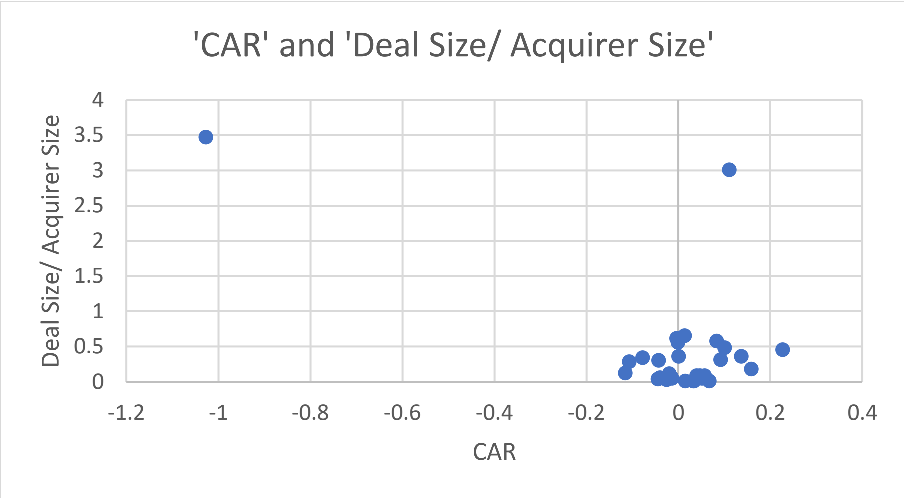


Table 6

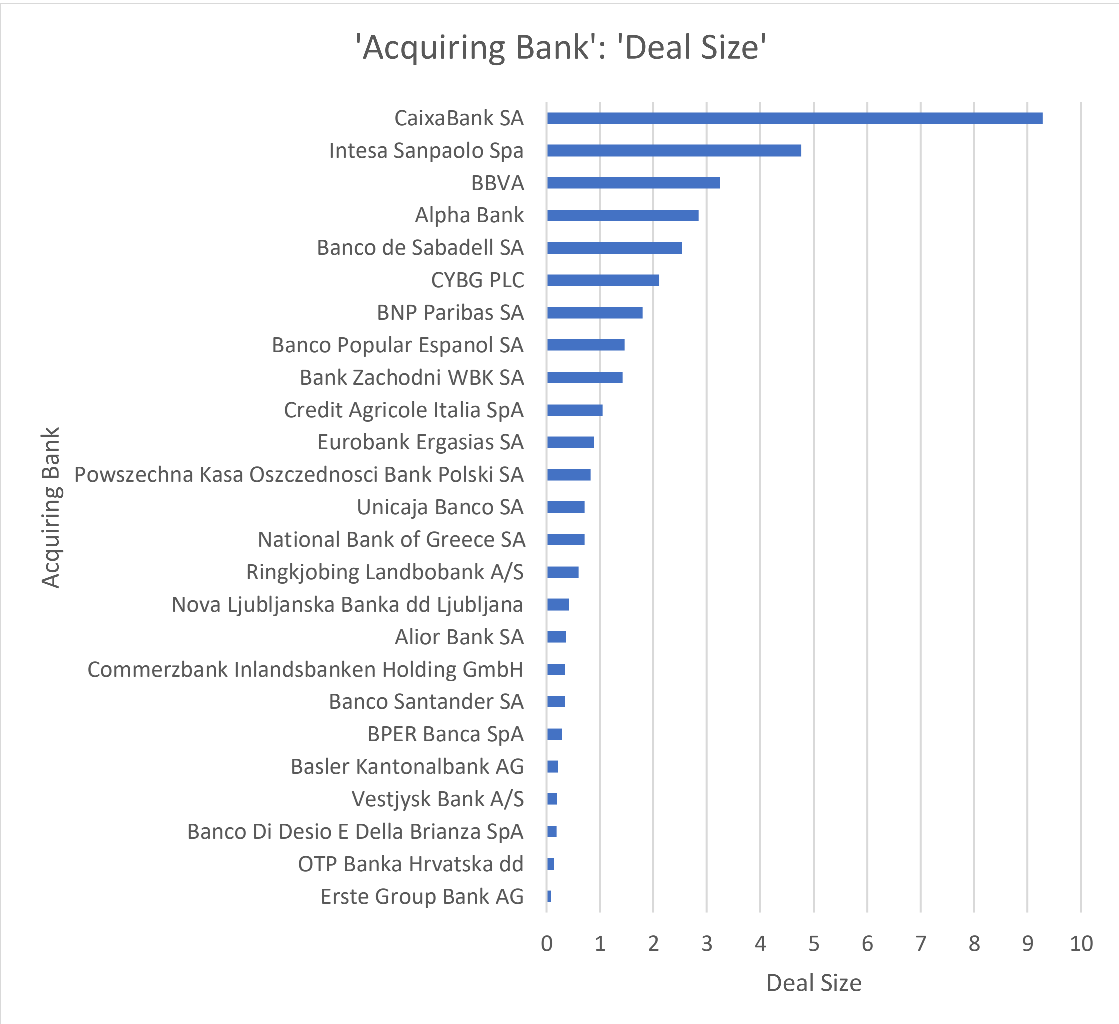


Table 7

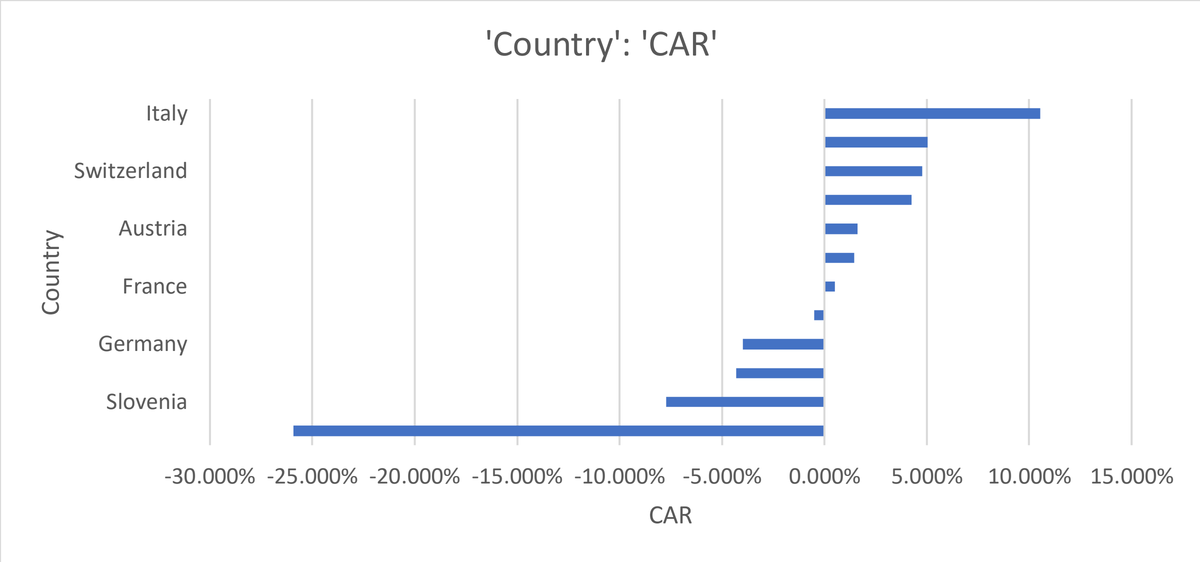


Table 8

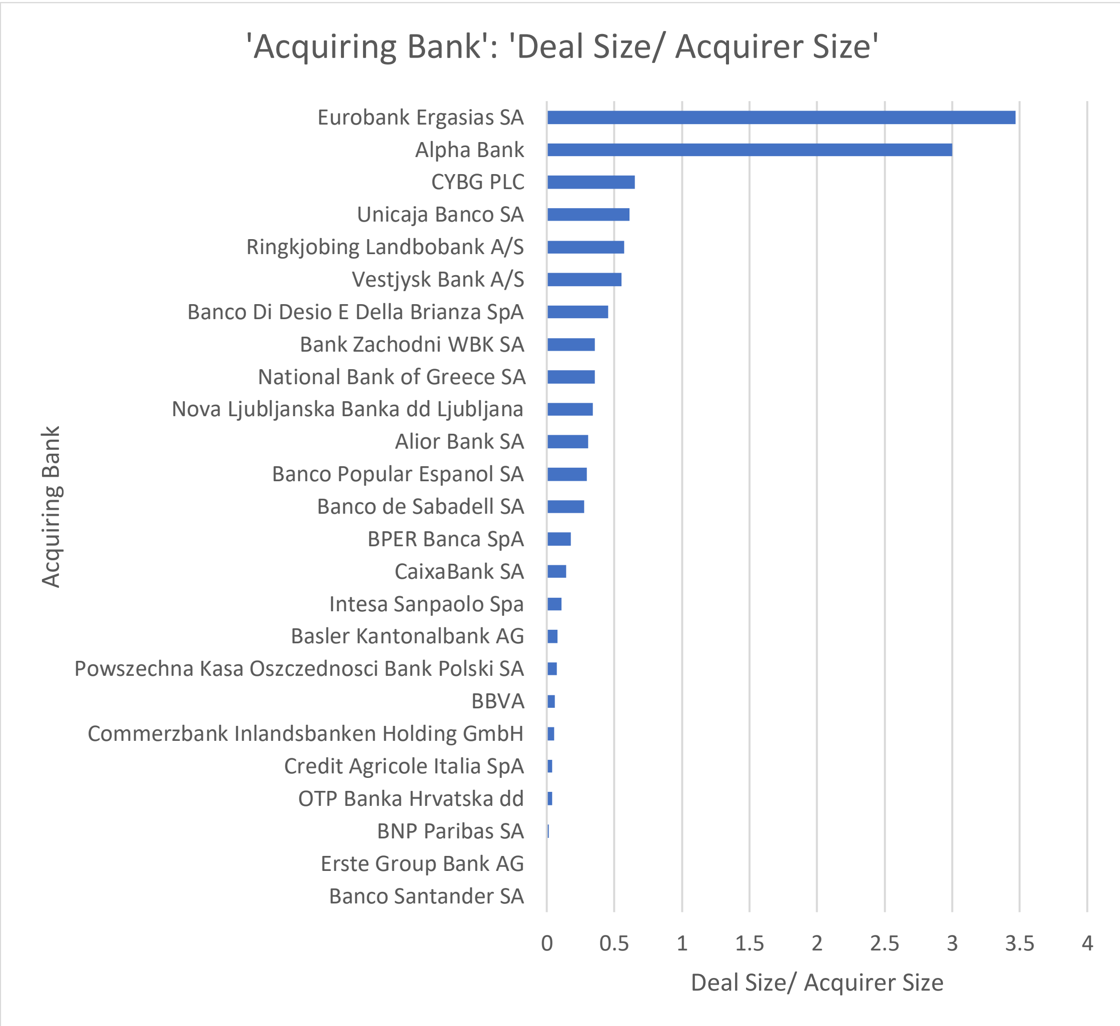


Table 9

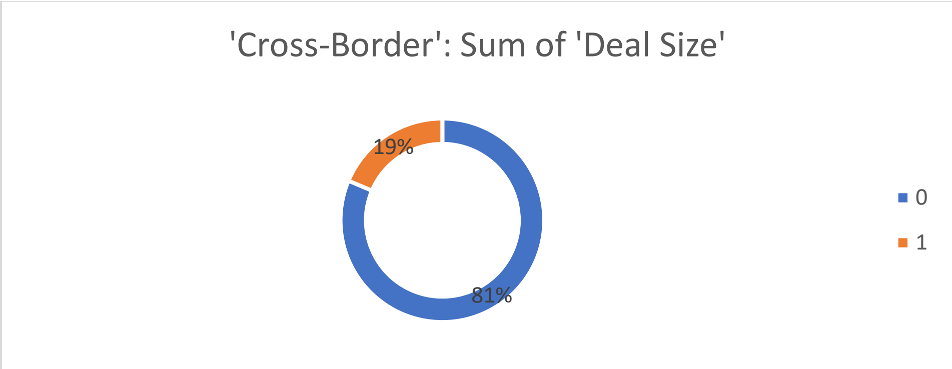


Table 10

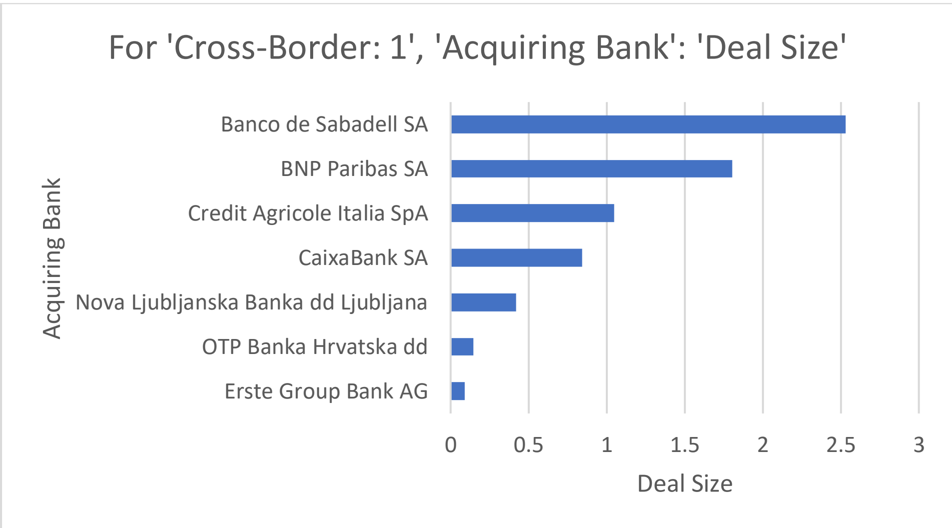


Table 11

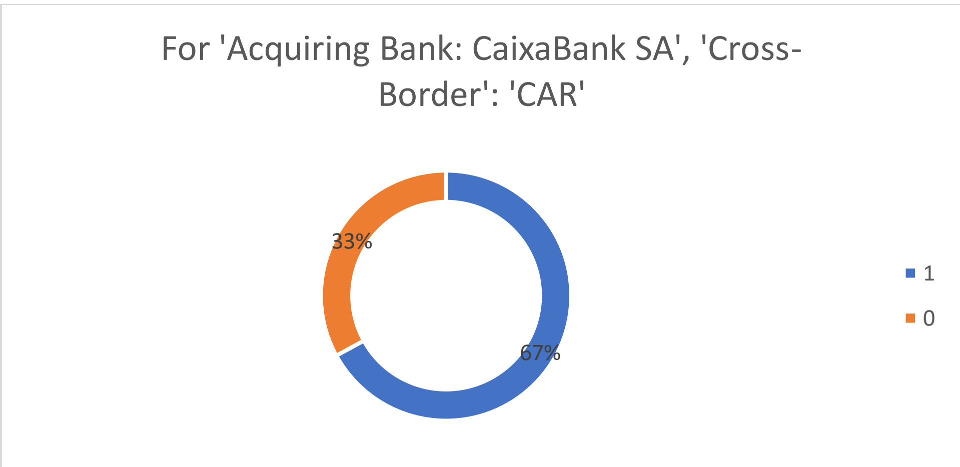


Table 12

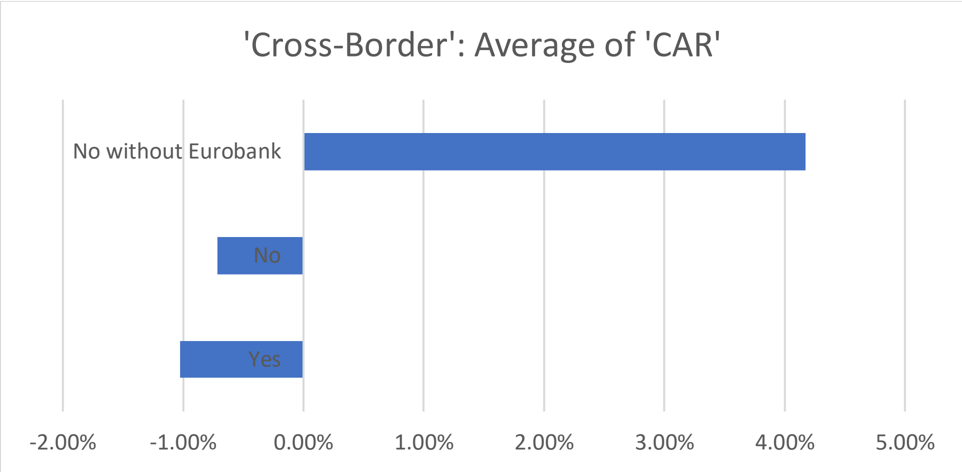


Table 13

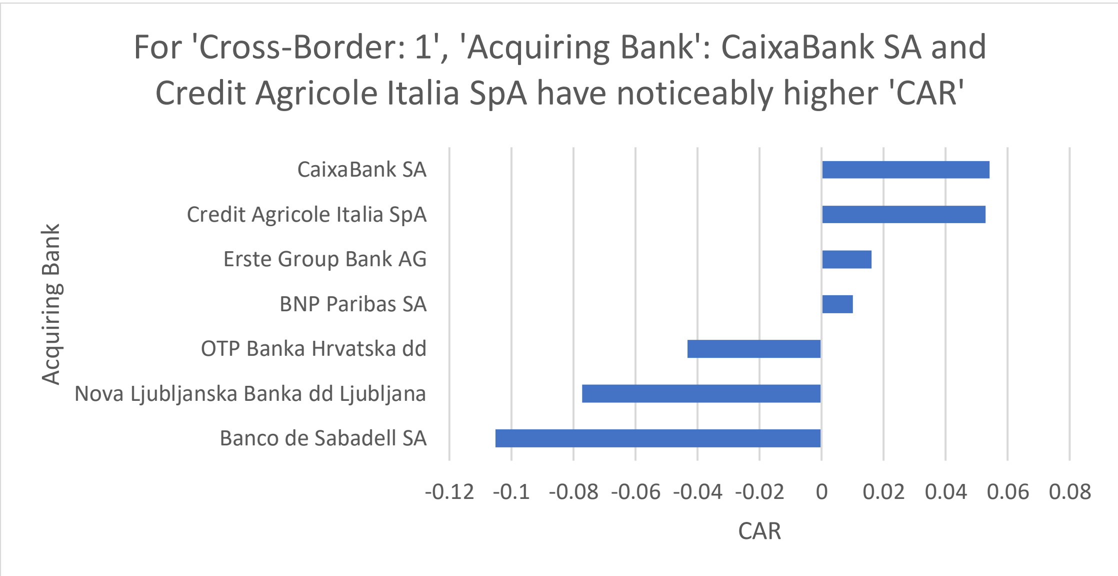


Table 14

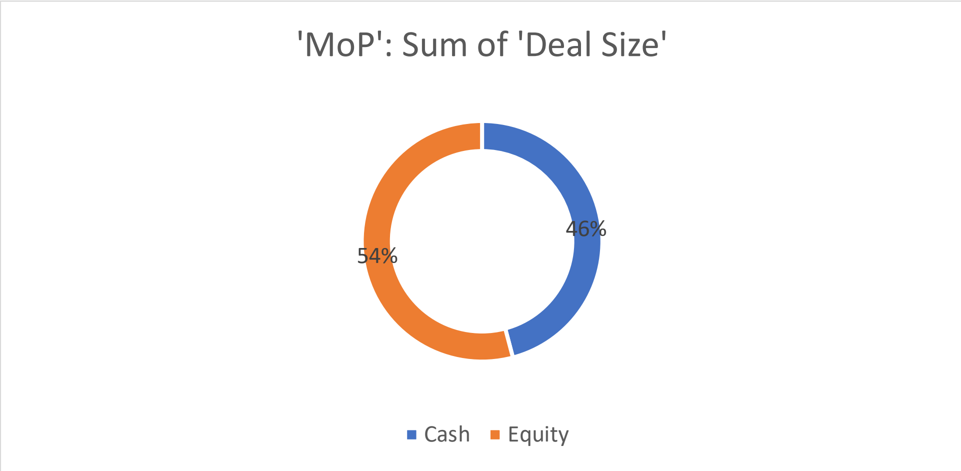


Table 15

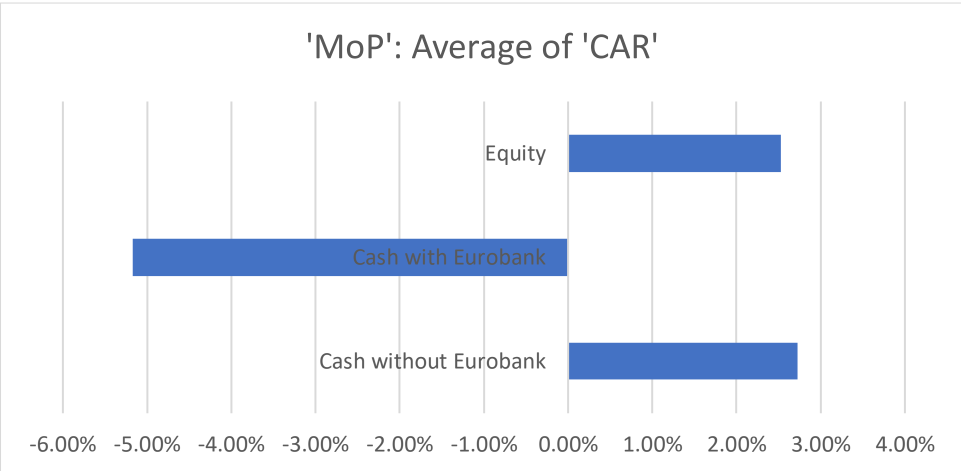


Table 16

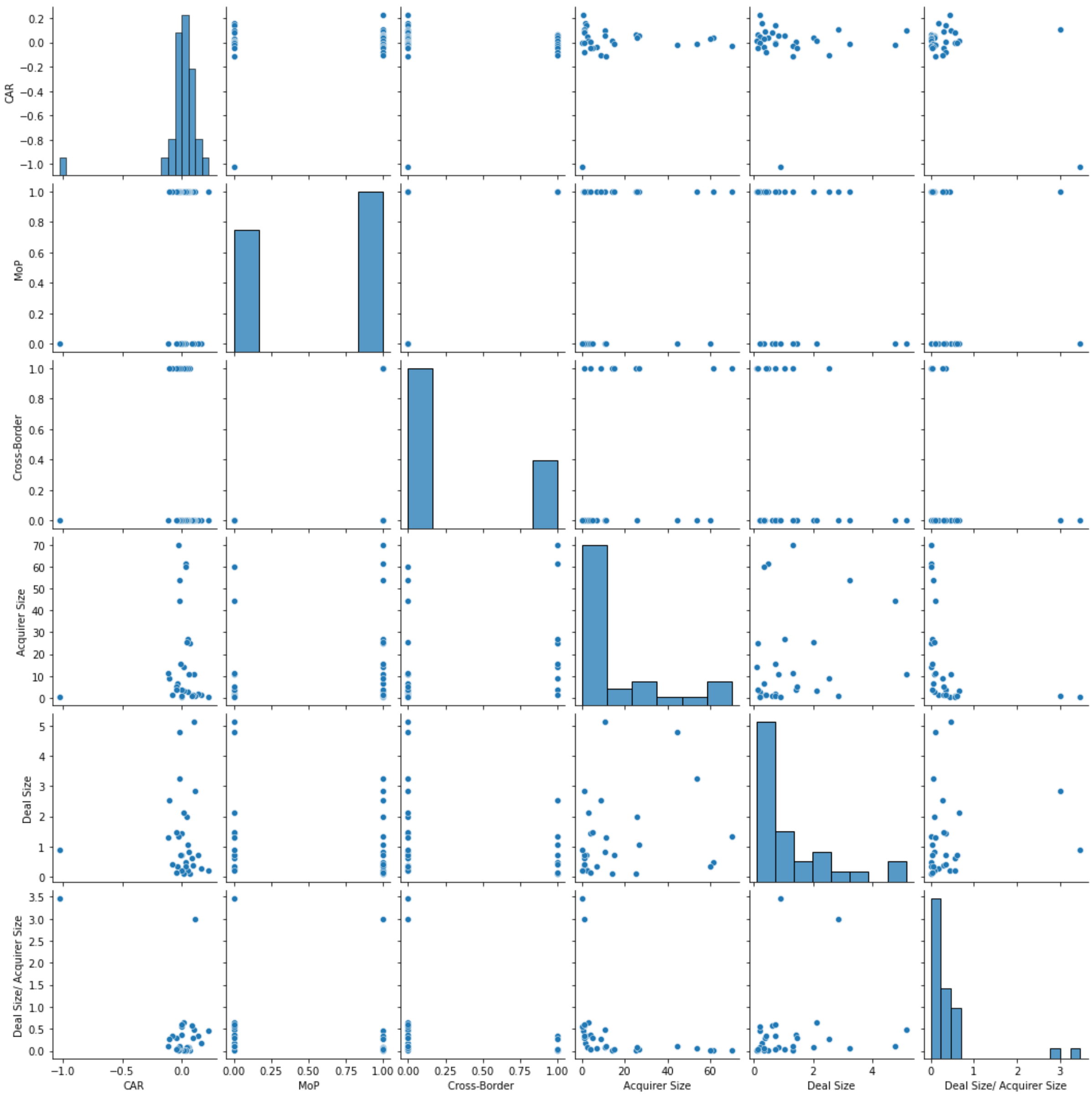
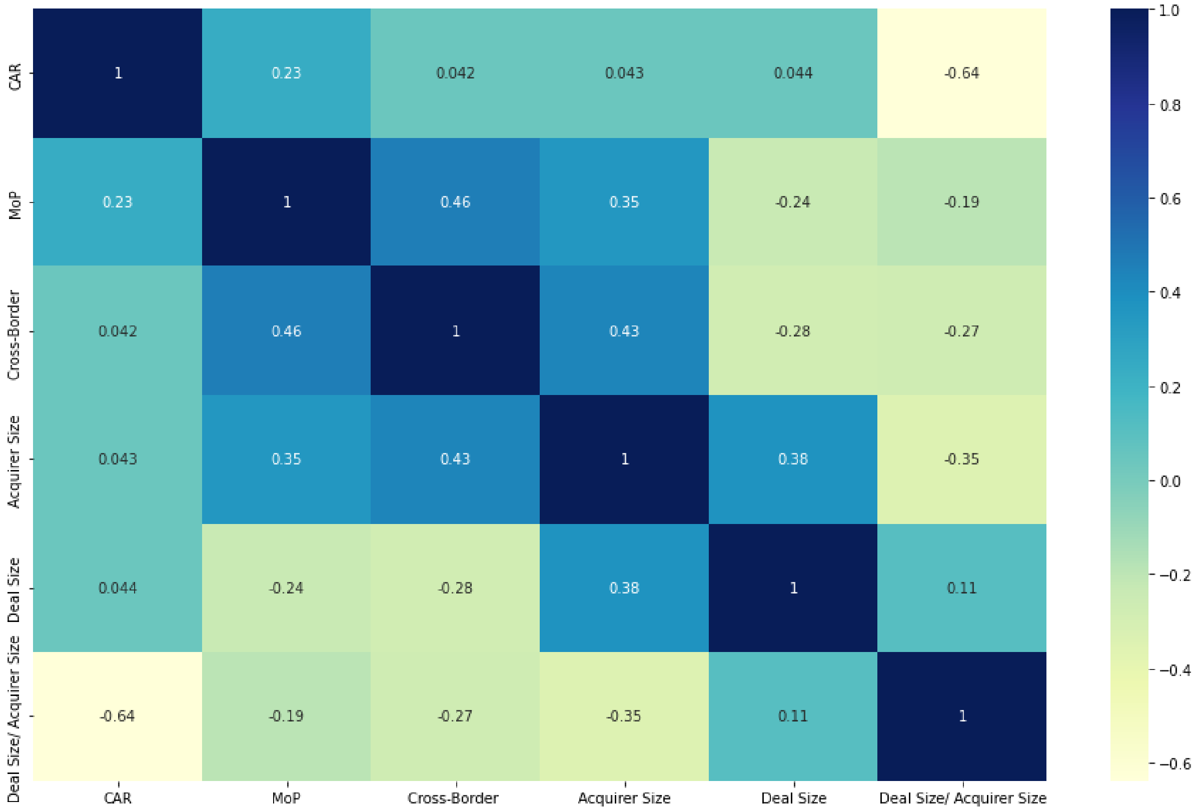

CAR
MoP
Cross-Border
Acquirer Size
Deal Size
Deal Size/ Acquirer Size

Table 17



CAR
MoP
Cross-Border
Acquirer Size
Deal Size
Deal Size/ Acquirer Size
1 0.23 0.042 0.043 0.044 -0.64
0.23 1 0.46 0.35 -0.24 -0.19
0.042 0.46 1 0.43 -0.28 -0.27
0.043 0.35 0.43 1 0.38 -0.35
0.044 -0.24 -0.28 0.38 1 0.11
-0.64 -0.19 -0.27 -0.35 0.11 1
CAR
MoP
Cross-Border
Acquirer Size
Deal Size
Deal Size/ Acquirer Size
1.0
0.8
0.6
0.4
0.2
0.0
−0.2
−0.4
−0.6